



Web Application Development II

Week 3: Bootstrap



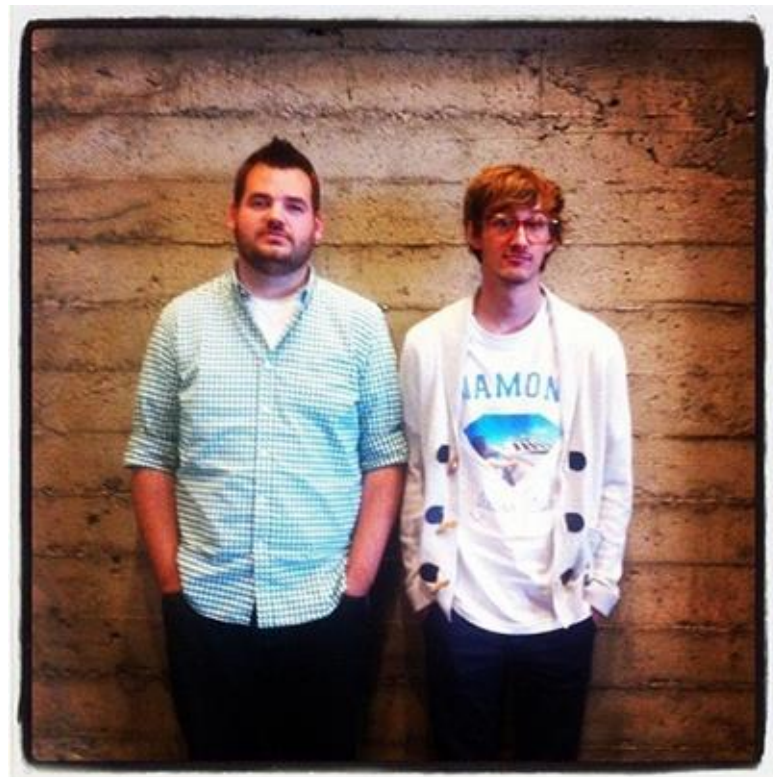
Agenda

What is Bootstrap?

How to use Bootstrap

Bootstrap examples

Build a project using Bootstrap



Mark Otto and Jacob Thornton

What is Bootstrap?

“The most popular HTML, CSS, and JS framework for developing responsive, mobile first projects on the web.”

Bootstrap is a giant collection of handy, reusable bits of code written in HTML, CSS, and JavaScript. It's a front-end development framework that enables developers to quickly build fully responsive websites.

Full details are at: www.getbootstrap.com

In programmers' language

Bootstrap =

`bootstrap.min.css + bootstrap.min.js`

What is Bootstrap?

Bootstrap highlights

Bootstrap's responsive grid

Bootstrap's Responsive Images

Bootstrap's Components

Bootstrap's JavaScript

...

Examples

<https://getbootstrap.com/docs/5.3/examples/>

Download the source code of the examples and customize accordingly if you wish.

Using Bootstrap

Using Bootstrap CDN

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>Bootstrap demo</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet" integrity="sha384-
QWTKZyjpPEjISv5WaRU9OFeRpok6YctnYmDr5pNlyT2bRjXh0JMhY6hW+ALEwIH" crossorigin="anonymous">
  </head>
  <body>
    <h1>Hello, world!</h1>
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js" integrity="sha384-
YvpcrYf0tY3IH B60NNkmXc5s9fDV ZLE SaAA55NDzOxhy9GkcldslK1eN7N6jleHz" crossorigin="anonymous"></script>
  </body>
</html>
```

Note: the latest Bootstrap version may not be "5.3.3". But it is okay as long as the Bootstrap version is "5.3.*"

This will load the libraries through the CDN - not from your server.

Using Bootstrap

Get started

<https://getbootstrap.com/docs/5.3/getting-started/introduction/>

The starter template

Refer to example: wk3example1.html

Exercise 1

- Open ex1.html (same as wk3example1.html) in your browser.
- Take note how the “Hello, world!” looks like.
- Comment out the Bootstrap CSS.
- See if the “Hello, world!” message looks different now.

“Use our powerful mobile-first flexbox grid to build layouts of all shapes and sizes thanks to a **twelve column** system, **six default responsive tiers**, Sass variables and mixins, and dozens of predefined classes.”

[illegible]

Bootstrap Features: Grid System

Responsive columns for easy layout and aligned content

Columns are centered within the parent .container

Each column has $1/n$ of the width

Bootstrap works with a 12 columns grid system

<https://getbootstrap.com/docs/5.3/layout/grid/>

Example:

wk3example2.html

Resize the browser to see the effect.
Change it to container-fluid and see the difference

Bootstrap Features: Grid System

Container: the most basic layout element in Bootstrap; required for using Bootstrap's grid system

Responsive breakpoints: scale up/down elements as the viewport (browser width) changes.

There are **six responsive breakpoints** – extra small (none), small (.sm), medium (.md), large (.lg), extra large (.xl) and extra extra large (.xxl).

Click [HERE](#) to find the value of the breakpoints.

Example

```
//resizes in chunks at breakpoints
<div class = "container">
    <!-- Content here -->
</div>
```

```
//continuously resizes as you change the
//width of your window/browser
<div class = "container-fluid">
    <!-- Content here -->
</div>
```

While containers *can* be nested, most layouts do not require a nested container.

Bootstrap Features: Grid System

Example

```
<style>
  .box {
    background-color: goldenrod;
    border: 1px solid black;
  }
</style>
```

Example: wk3example3.html

```
<div class="container">
  <div class="row">
    <div class="col box"> 1 of 3 </div>
    <div class="col-6 box"> 2 of 3 (wider)
  </div>
  <div class="col box"> 3 of 3</div>

</div>
</div>
```

1 of 3

2 of 3 (wider)

3 of 3

Bootstrap Features: Breakpoints

- customizable widths that determine how your responsive layout behaves across devices
- use different breakpoints to control when your layout can be adapted at a particular device size (viewport)

- **Example:** wk3example3.html, wk3example4.html

<!-- occupy 100% (fixed) width until 'lg' breakpoint is reached. After that, max-width is applied. This prevents the container from getting too large -->

```
<div class="container-lg">  
  content  
</div>
```

Breakpoints

Breakpoint	Class infix	Dimensions
Extra small	<i>None</i>	<576px
Small	sm	≥576px
Medium	md	≥768px
Large	lg	≥992px
Extra large	xl	≥1200px
Extra extra large	xxl	≥1400px

Bootstrap Features: Breakpoints

Use **.col-sm-*** classes to create a grid system that starts out stacked and becomes horizontal after reaching the small breakpoint (sm) onwards.

Example: *wk3example4.html*

```
<div class="container">  
  <div class="row">  
    <div class="col-sm-8">Lorem  
    Ipsum...</div>  
    <div class="col-sm-4">Lorem  
    Ipsum...</div>  
  </div>
```

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in



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At a small screen width

At a larger (than sm breakpoint) screen width

Grid Content Alignment

Place content in the center:

```
<div class="row justify-content-center">
```

Place content at the end:

```
<div class="row justify-content-end">
```

Place empty space equally around the content:

```
<div class="row justify-content-around">
```

Empty space in between the content:

```
<div class="row justify-content-between">
```

Exercise 2

Given ex2.html,

- referring to the various grid layouts based on the examples given in <https://getbootstrap.com/docs/5.3/layout/grid/>
- create a grid that looks similar to the one shown below.
- use class .container-fluid to span across the whole browser width.

1 of 2		2 of 2	
1 of 3	2 of 3 (wider)		3 of 3
1 of 4	2 of 4	3 of 4	3 of 4

Text

Use *text-center*, *text-end* classes to change the alignment of your text

Use *lead* class to make your paragraph stand out

Use *mark* class to highlight the text

<https://getbootstrap.com/docs/5.3/utilities/text/>

Example:

wk3example5.html

List

Use *list-unstyled* class to remove the bullet points

```

| Lorem ipsum dolor sit amet
| Consectetur adipiscing elit
| Integer molestie lorem at massa
```

Example

wk3example6.html

Use *list-inline* and *list-inline-item* classes to make the list inline

```

| Lorem ipsum  Phasellus iaculis  Nulla volutpat
```

Image

Make the image responsive (to browser width and resizing) by using *img-fluid* class and height: auto

Use *float-start*, *float-end* classes to align the images

Examples: *wk3example7.html*

```

```

```

```

```

```

Resize the browser to see the effect.



200x200



200x200

Utilities: Spacing

Bootstrap includes a wide range of shorthand responsive margin and padding utility classes to modify an element's appearance.

<https://getbootstrap.com/docs/5.3/utilities/spacing/>

m for margin and p for padding
x for left+right; y for top+bottom

The classes are named using the format {property}{sides}-{size} for xs and {property}{sides}-{breakpoint}-{size} for sm, md, lg, xl, and xxl.

Example:

mx-auto: margin on left and right auto (centered)

px-md-5: padding on left and right, breakpoint md (720px); size 5

Utilities: Display

Responsive display utility classes.

Classes can be combined for various effects as you need.

<https://getbootstrap.com/docs/5.3/utilities/display/>



Examples

Notation: *d-{value}* where value is one of: none, inline, inline-block, block, etc.

For example, display buttons inline, with padding size 2 and white-colored text:

```
<div class="d-inline p-2 bg-primary text-white">d-inline</div>
```

```
<div class="d-inline p-2 bg-dark text-white">d-inline</div>
```

Utilities: Sizing

Easily make an element as wide or as tall (relative to its parent) with sizing utilities.

<https://getbootstrap.com/docs/5.3/utilities/sizing/>



Examples

Support auto, 25%, 50%, 75%, 100% of the parent element

```
<div style="width: 400px; height: 200px; background-color: grey;">
```

```
  <div class="w-75 h-50 p-3" style="background-color: red;">
```

```
    Width 75%, Height 50% of the parent element</div>
```

```
</div>
```

Image Centering

Center the image using *mx-auto* and *d-block* classes

mx-auto class is for horizontally centering fixed-width block level content

Example: [wk3example7.html](#)

```

```



Table

In Bootstrap, table styling is opt-in i.e., you have to add `.table` class in the table tag to use Bootstrap's styling

Example: *wk3example8.html*

- Use `.table-dark` class to produce the dark colored theme
- Use `.table-borderless` class to remove the borders
- Use `.table-sm` class to make the table more compact
- Use `<caption>` tag to caption the table

Buttons

- Bootstrap includes several predefined button styles, each serving its own semantic purpose
- Use .btn-group to create Button toolbar horizontally or vertically
- The .btn classes can also be used with `<a>` or `<input>` elements

Example

wk3example9.html



```
<button type="button" class="btn btn-primary">Primary</button>
```



Exercise 3

Given ex3.html, search bootstrap documentation, and create a group of buttons arranged vertically - with your reasonable guess without Google first :-)



As a practice, also try to do the same thing (align the buttons vertically) using vanilla CSS.

Dropdown

To create dropdowns, wrap the dropdown's toggle (button or link) and the dropdown menu within `.dropdown`

Use `.dropdown-toggle` and `data-toggle="dropdown"` attribute to link the toggle to the `.dropdown-menu`

(so that it is there once clicked; and it is not there once clicked again)

Examples

wk3example10.html

Collapse

Collapse class indicates a collapsible element (the content that will be shown or hidden with a click of a button).

To control (show/hide) the collapsible content, add the `data-toggle="collapse"` attribute to an `<a>` or a `<button>` element.

Then add the `target="#id"` attribute to link.

<https://getbootstrap.com/docs/5.3/components/collapse/>

Examples

wk3example11.html

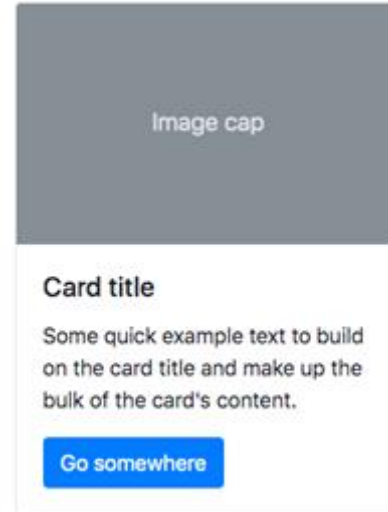
Card

Bootstrap provides a ready-to-use card that can contain image and text

<https://getbootstrap.com/docs/5.3/components/card/>

Examples

wk3example12.html



Carousel

The carousel is a slideshow for cycling through a series of content, built with CSS 3D transforms and a bit of JavaScript. It works with a series of images, text, or custom markup. It also includes support for previous/next controls and indicators.

<https://getbootstrap.com/docs/5.3/components/carousel/>

Examples

wk3example13.html



Carousel: Cont'd

- The `.active` class needs to be added to one of the slides otherwise the carousel will not be visible.
- set a unique id on the `.carousel` for optional controls
- Control and indicator elements must have a `data-target` attribute (or `href` for links) that matches the id of the `.carousel` element.

Examples

wk3example13.html

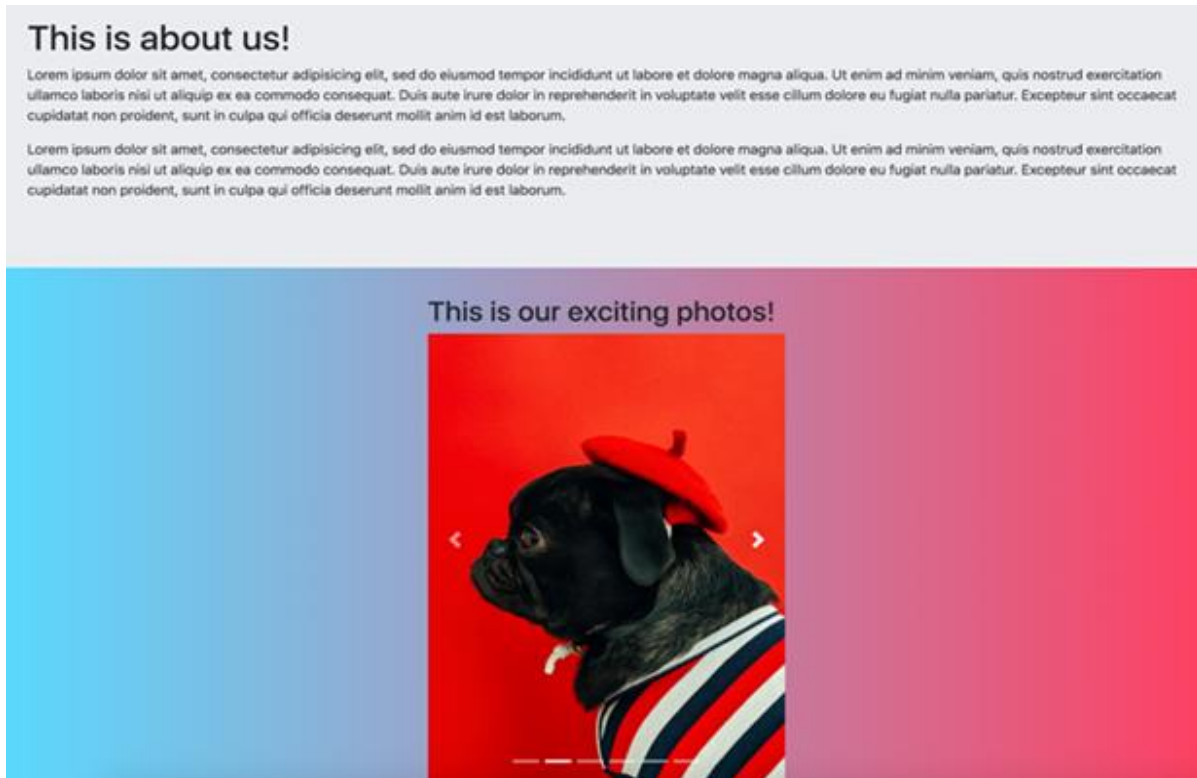


Exercise 4

Add code in ex4-about.html to create a web page that looks like this.

Hint: for the top part, consider jumbotron
(*wk3example14.html*)

Hint: consider carousel.



Form

Bootstrap provides various form control styles, layout options, and custom components for creating forms.

Use appropriate type attribute on all inputs to take advantage of newer input controls like email verification and number selection.

<https://getbootstrap.com/docs/5.3/forms/overview/>

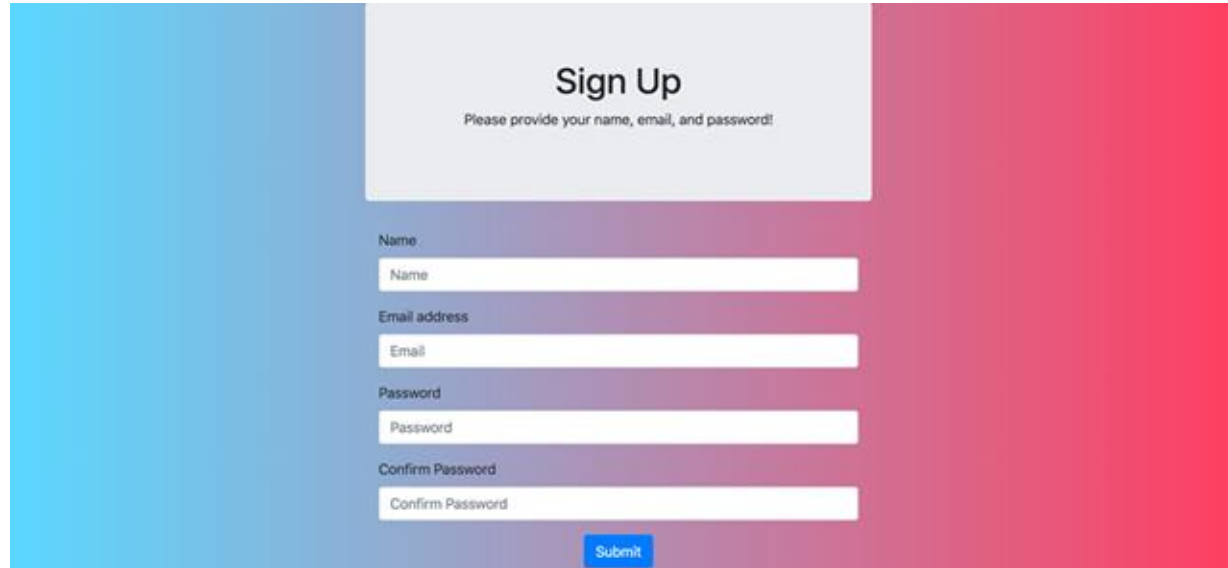
Example

wk3example15.html

Exercise 5

Add code in ex5-signup.html to create a sign up form that looks like this.

Hint: form



Sign Up

Please provide your name, email, and password!

Name

Email address

Password

Confirm Password

Navs

Bootstrap provides ready-to-use, flexible, well-styled navigation links and bars.

<https://getbootstrap.com/docs/5.3/components/navs-tabs/>

Example: *wk3example16.html*

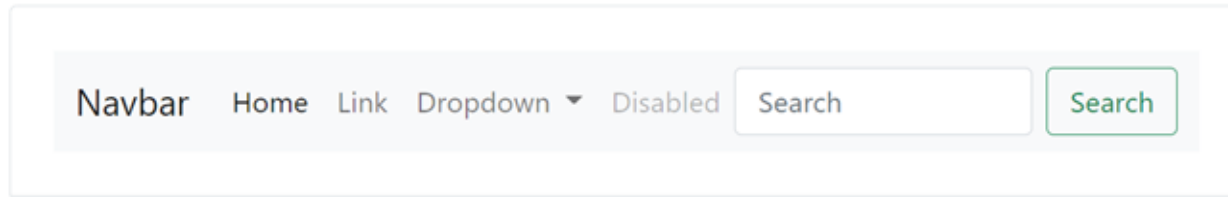
```
<ul class="nav">
  <li class="nav-item">
    <a class="nav-link" href="/home.php">Home</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="/login.php">Log In</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="/logout.php">Log Out</a>
  </li>
</ul>
```

[Home](#)[Log In](#)[Log Out](#)

NavBar

Bootstrap's powerful, responsive navigation header, the navbar. **Example:** *wk3example17.html*

<https://getbootstrap.com/docs/5.3/components/navbar/>



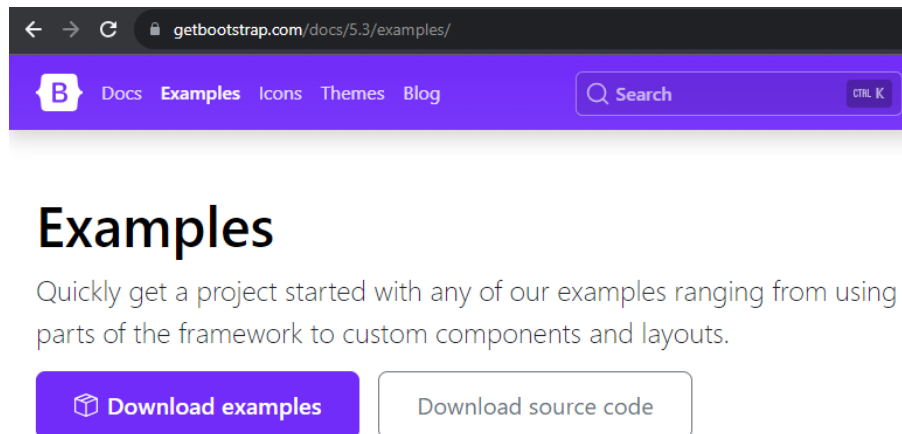
Extra: Accessible Rich Internet Applications Suite

<https://www.w3.org/TR/wai-aria/>

- WAI-ARIA is a spec defining support for making web content more accessible to people with disabilities.
- It defines bunch of markup extensions (mostly as attributes on HTML5 elements), which can be used by the web app developer to provide additional information about the semantics of the various elements to assistive technologies like screen readers.
- Browser interprets the ARIA markup with best effort, but the spec is created in such a way that ignoring the ARIA-specific markup will not affect the web app's functionality.

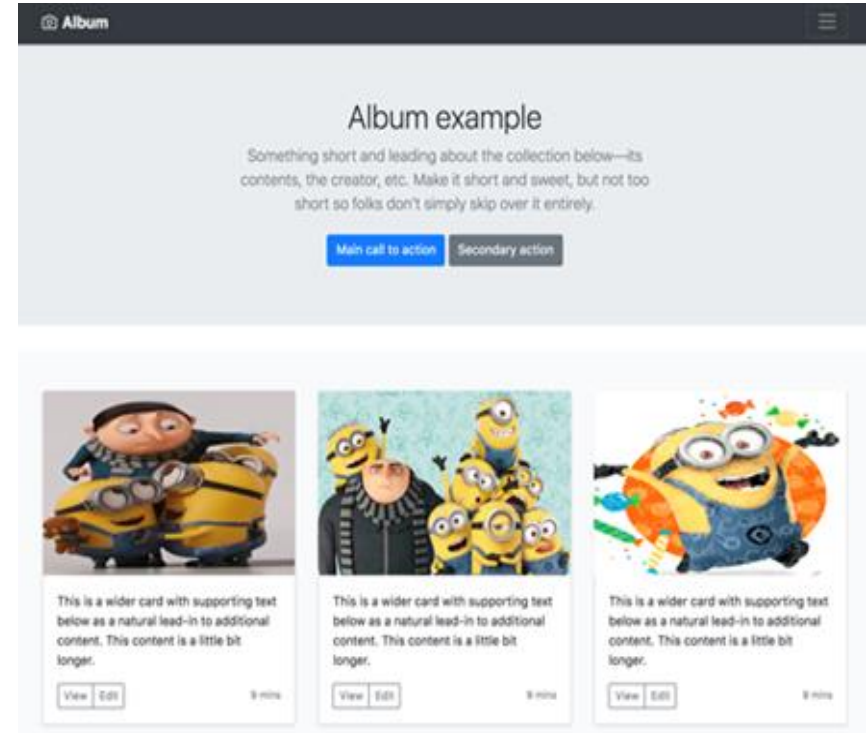
Bootstrap Templates

- Go to: <https://getbootstrap.com/docs/5.3/examples/>
- Download examples ([LINK](#))
- Unzip the folder
- Customize bootstrap examples



Exercise 6: Album

- Unzip the folder
- Go to the unzipped folder
- Go to the sub-folder “album” (ref: <https://getbootstrap.com/docs/5.3/examples/album/>)
- Copy the sample album code into ex6 folder and create your own album page



Take Away Messages

- Bootstrap pre-defines many classes.
- Use the grid system to have a responsive layout.